

Global Optimizations

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Types of optimizations

Classification of optimization (based on their scope)

- Local (within basic blocks)
- Global Intra-procedural
- Global Inter-procedural

Classification based on their positioning:

- High level optimizations (use the program structure to optimize).
- Low level optimizations (work on medium/lower level IR)

Optimization classification (contd)

Classification with respect to their dependence on the target machine.

Machine independent

- applicable across broad range of machines
- Examples
 - move evaluation to a less frequently executed place
 - remove redundant (unreachable, useless) code.
- create opportunities.

Machine dependent

- capitalize on machine-specific properties
- improve mapping from IR onto machine
- strength reduction.
- replace sequence of instructions with more powerful one (use “exotic” instructions)

Desirable properties of an optimizing compiler

- code at least as good as an assembler programmer
- stable, robust performance (predictability)
- architectural strengths fully exploited
- architectural weaknesses fully hidden
- broad, efficient support for language features
- instantaneous compilation

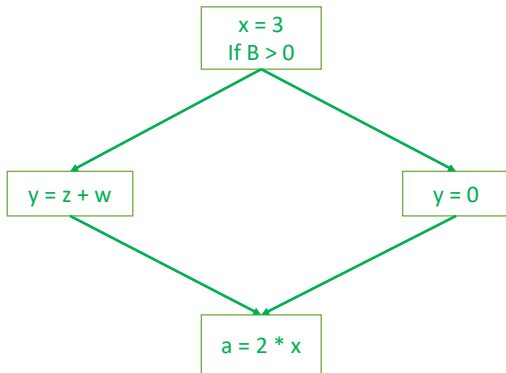
From local to global optimization

- Recall the local basic-block optimizations
 - Constant propagation
 - Dead code elimination

$x = 3$		$x = 3$		$y = z * w$
$y = z * w$	$\rightarrow$	$y = z * w$	$\rightarrow$	$q = 3 + y$
$q = x + y$		$q = 3 + y$		

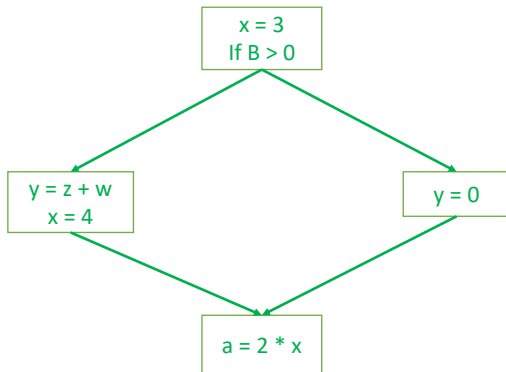
From local to global optimization

Can these optimizations be directly extended to an entire control-flow graph?



From local to global optimization

Can these optimizations be directly extended to an entire control-flow graph? There are situations where it is incorrect to globally propagate constants:



From local to global optimization: Constant Propagation

Correctness Criterion

To replace a use of x by constant k , on every path to the use of x , the last assignment to x is $x = k$.

- This correctness criterion is non-trivial to check.
- ‘Every path’ includes paths around loops and through branches of conditionals.
- This requires a ‘global’ analysis, i.e. an analysis of the entire control-flow graph.

Global Analysis

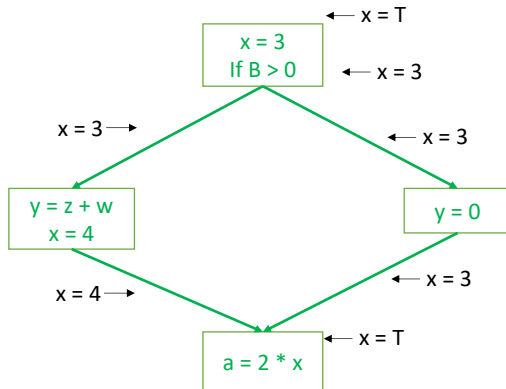
Global optimization tasks share several traits:

- The optimization depends on knowing some property X at a particular point in program execution.
- Proving X at any point requires knowledge of the entire function.
- It is OK to be conservative. If the optimization requires X to be true, then the analysis can output one of two things:
 - X is definitely true
 - Don't know if X is true.
- Global dataflow analysis is a standard technique for performing global optimizations.

Global Constant Propagation

- To perform global constant propagation at a program point, we need to know whether a variable will always have a constant value at the point.
- We associate one of the following values with variable x at every program point:
 - $\perp$: This program point is not reachable.
 - c : x has a constant value c
 - $\top$: x is not a constant

Example



Defining Global Constant Propagation Analysis

- We associate two functions $in, out : Var \rightarrow \mathbb{Z} \cup \{\top, \perp\}$ with each basic block.
- Global Constant Propagation Analysis is a forward analysis: in of a basic block is defined in terms of out of predecessors.
- For a basic block b , In_b is defined as follows:

$$in_b(x) = \begin{cases} \top & \exists p \in Pred(b). out_p(x) = \top \\ c & \forall p \in Pred(b). out_p(x) = c \vee out_p(x) = \perp \\ \top & \exists p_1, p_2 \in Pred(b). out_{p_1}(x) \neq out_{p_2}(x) \\ \perp & \forall p \in Pred(b). out_p(x) = \perp \end{cases}$$

The above definition is also called a meet operation.

We use the following notation for the above definition:

$$in_b = \bigvee_{p \in Pred(b)} out_p$$

Defining Global Constant Propagation Analysis

out_b is defined in terms of in_b and the statements in basic block b
For simplicity, assume that we have a separate basic block for each statement:

$$out_b(x) = \begin{cases} \perp & in_b(x) = \perp \\ c & b : x = c \text{ where } c \in \mathbb{Z} \\ \top & b : x = e \text{ where } e \text{ is an expression} \\ In_b(x) & b : y = \dots \end{cases}$$

The above definition is also called a transfer function.

We can express the definition as a function f_b such that $f_b(in_b) = out_b$.

Defining Global Constant Propagation Analysis

$$out_b(x) = \begin{cases} \perp & in_b(x) = \perp \\ c & b : x = c \text{ where } c \in \mathbb{Z} \\ e[In_b] & b : x = e \text{ where } e \text{ is an expression} \\ In_b(x) & b : y = \dots \end{cases}$$

We can also do constant folding while evaluating expressions.

Given a function $f : Var \rightarrow \mathbb{Z} \cup \{\top, \perp\}$, $e[f]$ denotes the evaluation of expression e using function f .

While evaluating, $\top + c = \top$ (similar for other arithmetic operators).

Iterative method for computing *In*, *Out*

N : Set of nodes of CFG;

$Start$: Entry basic blocks of CFG (i.e. successors of `entry`);

foreach $\underline{n \in Start}$ **do**

$in_n \leftarrow \lambda v. \top$;

end

foreach $\underline{n \in N - Start}$ **do**

$in_n \leftarrow \lambda v. \perp$;

$out_n \leftarrow \lambda v. \perp$;

end

repeat

foreach $\underline{n \in Nodes}$ **do**

$in'_n \leftarrow in_n$;

$out'_n \leftarrow out_n$;

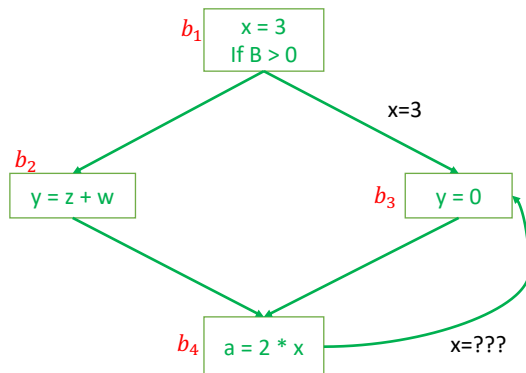
$in_n \leftarrow \bigvee_{p \in Pred(n)} out_p$;

$out_n \leftarrow f_n(in_n)$;

end

until $\underline{\forall n, in'_n = in_n \wedge out'_n = out_n}$;

Why do we need $\perp$?



- To compute in_{b_3} , we need out_{b_4} , but for that we need out_{b_3} !
- We will encounter similar problems whenever we have loops.
- Hence, we initialize in and out values with $\perp$, which intuitively means that ‘so far as we know, control never reaches this point’

Orderings

- We can simplify the presentation of the data-flow analysis by ordering the values.
 - $\forall c \in \mathbb{Z}. \top \leq c \leq \perp$
 - Formally, $\leq$ is a partial order on the set $\mathbb{Z} \cup \{\perp, \top\}$.
- We can define the greatest lower bound of a set of values.
 - Formally, $(\mathbb{Z} \cup \{\perp, \top\}, \leq)$ forms a meet semi-lattice, and hence the *glb* always exists for any set of values.

$$in_b(x) = \begin{cases} \top & \exists p \in Pred(b). out_p(x) = \top \\ \top & \exists p_1, p_2 \in Pred(b). out_{p_1}(x) \neq out_{p_2}(x) \\ \perp & \forall p \in Pred(b). out_p(x) = \perp \\ c & \forall p \in Pred(b). out_p(x) = c \vee out_p(x) = \perp \end{cases}$$

- Notice that $in_b(x) = glb(\{out_p(x) \mid p \in Pred(b)\})$.
 - The greatest lower bound is also called **meet**.

Orderings

- Every data-flow analysis can be represented by defining its meet semi-lattice, with the corresponding meet operation being used in the iterative method.
 - Useful for proving the soundness of the analysis, for comparing precision of different analyses, and for proving termination of the iterative method.
 - Dataflow analysis/Abstract Interpretation covered in detail in advanced courses: CS5030, CS6013.
- Termination argument: We start with the highest value ($\perp$) and we only move down.
 - $\perp$ can change to a constant value, which can change to $\top$.
 - Thus, each $in(x)$ or $out(x)$ can change at most twice at any basic block.
 - Maximum number of iterations = $2 * 2 * \text{Number of variables} * \text{Number of basic blocks}$.

Liveness Analysis... revisited

- We can represent liveness analysis in the dataflow analysis framework.
- Let Var be the set of variables. Then, the meet semi-lattice would be $(\mathbb{P}(Var), \supseteq)$.
 - The glb operation is set union.
 - The analysis works in the backward direction. Hence,
$$out_b = \bigcup_{s \in succ(b)} in_s.$$
- The transfer function is $f_b(S) = use_b \cup (S - def_b)$.
 - use_b are variables which are used before they are (possibly) defined in b . Can be determined using the next-use algorithm.
 - def_b are variables which are defined in b .

Types of program analysis

Classification of analysis (based on their view)

```
if (cond) {  
    a = ...  
    b = ...  
} else {  
    a = ...  
    c = ...  
}  
  
// Which of the variables may be assigned? -- {a,b,c}  
// Which of the variables must be assigned? -- {a}
```

- May analysis – the analysis holds on at least one data flow path.
- Must analysis – the analysis must hold on all data flow paths.
 - What can we say about constant propagation analysis? May or Must?
 - What can we say about liveness analysis? May or Must?

Classification of analysis (contd)

Classification of analysis (based on precision)

- Flow sensitive / insensitive.
 - Insensitive - the analysis should hold at every program point; does not depend on the control flow.
 - Sensitive - Each program point has its own analysis.

```
if (c) {  
    a = 2;  
    b = a;  
    c = 3;  
    print (a, b, c); // constants?  
} else {  
    a = 3  
    b = a;  
    c = 3;  
    print (a, b, c); // constants?  
}
```

Classification of analysis (contd)

- Context sensitive and insensitive

```
a = foo(2);
```

```
b = foo (3);
```

```
c = bar (2);
```

```
d = bar(2);
```

```
print (a, b, c, d); // a, b, c, d constants?
```

```
int foo(int x) { return x }
```

```
int bar(int x) { return x * x }
```

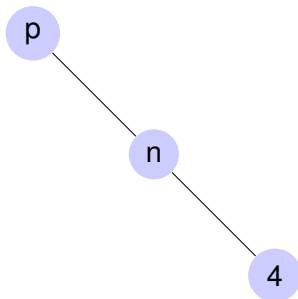
Alias Analysis

Alias analysis: problem of identifying storage locations that can be accessed by more than one way.

Are variable `a` and `b` aliases? $\Rightarrow$ `a` and `b` refer to the same location?
Modifying the contents of `a`, modifies the contents of `b`.

Necessary for performing many optimizations such as constant/copy propagation, common sub-expression elimination, dead code elimination, etc.

```
foo() {  
    int *p;  
    int n;  
    p = &n;  
    n = 4;  
    print ("%d", *p);  
}
```



Alias analysis (contd)

```
extern int *q;  
foo( ) {  
    int a = 0, k;  
    k = a + 5;  
    f (a, &k);  
    *q = 13;  
    k = a + 5;    /* Assignment is redundant? */  
                  /* Expression is redundant? */  
    ...  
}
```

What happens if $q == k$?

Loop unrolling

(Example) Matrix-matrix multiply

```
do i ← 1, n, 1
  do j ← 1, n, 1
    c(i, j) ← 0
    do k ← 1, n, 1
      c(i, j) ← c(i, j) + a(i, k) * b(k, j)
```

- All the array elements are floating point values.
- $2n^3$ flops, n^3 loop increments and branches
- each iteration does 3 loads and 2 flops

Example: loop unrolling

Matrix-matrix multiply

(assume 4-word cache line)

```
do i ← 1, n, 1
  do j ← 1, n, 1
    c(i, j) ← 0
    do k ← 1, n, 4
      c(i, j) ← c(i, j) + a(i, k) * b(k, j)
      c(i, j) ← c(i, j) + a(i, k+1) * b(k+1, j)
      c(i, j) ← c(i, j) + a(i, k+2) * b(k+2, j)
      c(i, j) ← c(i, j) + a(i, k+3) * b(k+3, j)
```

- $2n^3$ flops, $\frac{n^3}{4}$ loop increments and branches
- each iteration does 9 loads and 8 flops
- memory traffic is better
 - $c(i, j)$ is reused
 - $a(i, k+...)$ reference are from cache
 - $b(k, j)$ is problematic

(put it in a register)

Example: loop unrolling

Matrix-matrix multiply

(to improve traffic on b)

```
do j ← 1, n, 1
  do i ← 1, n, 4
    c(i,j) ← 0
    do k ← 1, n, 4
      c(i,j) ← c(i,j) + a(i,k) * b(k,j)
        + a(i,k+1) * b(k+1,j) + a(i,k+2) * b(k+2,j)
        + a(i,k+3) * b(k+3,j)
      c(i+1,j) ← c(i+1,j) + a(i+1,k) * b(k,j)
        + a(i+1,k+1) * b(k+1,j)
        + a(i+1,k+2) * b(k+2,j)
        + a(i+1,k+3) * b(k+3,j)
      c(i+2,j) ← c(i+2,j) + a(i+2,k) * b(k,j)
        + a(i+2,k+1) * b(k+1,j)
        + a(i+2,k+2) * b(k+2,j)
        + a(i+2,k+3) * b(k+3,j)
      c(i+3,j) ← c(i+3,j) + a(i+3,k) * b(k,j)
        + a(i+3,k+1) * b(k+1,j)
        + a(i+3,k+2) * b(k+2,j)
        + a(i+3,k+3) * b(k+3,j)
```

Example: loop unrolling

What happened?

- interchanged i and j loops
- unrolled i loop
- fused inner loops
- $2n^3$ flops, $\frac{n^3}{16}$ loop increments and branches
- first assignment does 9 loads and 8 flops
- 2nd through 4th do 5 loads and 8 flops
- memory traffic is better
 - $c(i+..., j)$ is shared across 4 iterations w.r.t the original program
 - $a(i+..., k+...)$ references are from cache
 - $b(k+..., j)$ is reused

(register)

Loop optimizations: factoring loop-invariants

Loop invariants: expressions constant within loop body

Goal: move the loop invariant computation to outside the loop.

The loop independent code executes only once, instead of many times the loop might.

Example: loop invariants

```
foreach i=1 .. 100 do
  foreach j=1 .. 100 do
    foreach k=1 .. 100 do
      | A[i, j, k] = i * j * k;
    end
  end
end
```

- 3 million index operations
- 2 million multiplications

Example: loop invariants (cont.)

Factoring the inner loop:

```
foreach i=1 .. 100 do  
  | foreach j=1 .. 100 do  
  | | t1 = &A[i][j];  
  | | t2 = i * j ;  
  | | foreach k=1 .. 100 do  
  | | | t1[k] = t2 * k;  
  | | end  
  | end  
end
```

And the second loop:

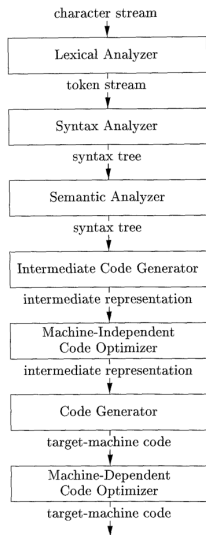
```
foreach i=1 .. 100 do  
  | t3 = &A[i];  
  | foreach j=1 .. 100 do  
  | | t1 = &t3[j];  
  | | t2 = i * j ;  
  | | foreach k=1 .. 100 do  
  | | | t1[k] = t2 * k;  
  | | end  
  | end  
end
```

Compilers are engineered objects

- minimize running time of compiled code
- minimize compile time
- use reasonable compile-time space
- find a reasonable trade-off

Thus, results are sometimes unexpected

Back to first lecture



Front end responsibilities:

- Recognize syntactically legal code; report errors.
- Recognize semantically legal code; report errors.
- Produce IR.

Back end responsibilities:

- Optimizations, code generation.

Our target

- five out of seven phases.
- glance over optimizations.